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Course:Wireless Network

Section: BCS-6(B)

ASSIGNMENT 1

Simulation and Performance Analysis of Medium Access Control (MAC) Protocols

Introduction

In modern communication systems, many users need to send data using the same communication channel. Since the channel is shared, there must be a proper way to control how users access it. This is done using Medium Access Control (MAC) protocols.

MAC protocols decide when a user can transmit data so that collisions (when multiple users send data at the same time) can be reduced and the channel can be used efficiently.

There are two main types of MAC protocols. The first type is random access protocols, such as Pure ALOHA, Slotted ALOHA, CSMA/CD, and CSMA/CA. In these protocols, users try to send data randomly, which can cause collisions, but different techniques are used to reduce them.

The second type is controlled access protocols, such as TDMA, FDMA, and CDMA. In these protocols, the channel is divided among users using time, frequency, or codes, which helps to avoid collisions.

In this assignment, a simulation is developed in Python to study the behavior of different MAC protocols. The simulation models multiple users trying to access a shared channel and compares their performance using different metrics like throughput, collision rate, delay, and fairness.

Methodology

In this assignment, the simulation of MAC protocols is implemented using Python. The main goal is to model multiple users trying to access a shared communication channel and observe how different protocols behave.

Simulation Design

The simulation works in discrete time steps. At each time step, every user decides randomly whether to transmit a packet based on a given transmission probability.

For each protocol, a separate logic is implemented:

In **Pure ALOHA**, users transmit at any time without checking the channel, which leads to a high number of collisions.

In **Slotted ALOHA**, transmissions are allowed only at specific time slots, which reduces collisions.

In **CSMA/CD**, users first check if the channel is busy before transmitting. If a collision occurs, it is detected and handled.

In **CSMA/CA**, collisions are avoided using a backoff mechanism, where users wait for a random time before retransmitting.

In **TDMA**, each user is assigned a fixed time slot, so only one user transmits at a time.

In **FDMA**, each user is given a separate frequency, allowing multiple users to transmit simultaneously without collision.

In **CDMA**, multiple users can transmit at the same time, but a simplified model is used where performance decreases if too many users transmit together.

During the simulation, the following values are tracked:

- Number of successful transmissions
- Number of collisions
- Total transmission attempts

- Delay experienced by packets
- Successful transmissions per user (for fairness)

These values are then used to calculate performance metrics such as throughput, collision rate, channel utilization, average delay, and fairness.

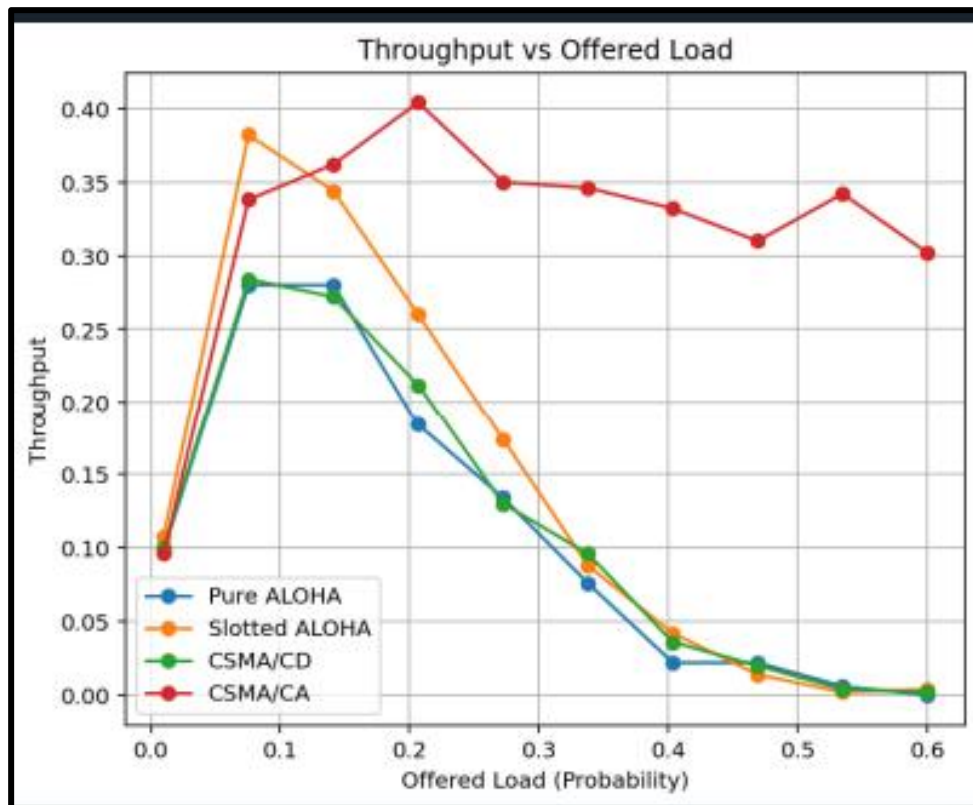
Assumptions

The simulation is based on the following assumptions:

- Time is divided into equal discrete steps
- All users have the same probability of transmitting
- The communication channel is shared among all users
- Collisions occur when more than one user transmits at the same time (for random access protocols)
- No real-world factors like noise, signal loss, or propagation delay are considered
- CSMA protocols use a simplified channel sensing model
- CSMA/CA uses a random backoff time for retransmission
- TDMA assigns equal time slots to all users
- FDMA divides bandwidth equally among users
- CDMA is implemented as a simplified model with a limit on simultaneous users (does not include near–far effect)

Results

Graph 1: Throughput vs Offered Load



Observations:

General Pattern: At first, as more traffic (Load) is added, the throughput goes up. However, after a certain point, the throughput drops because too many packets are crashing into each other (collisions).

Pure ALOHA (Blue Line): This protocol has the **lowest performance**. It fails quickly as traffic increases because it has no rules to prevent collisions.

Slotted ALOHA (Orange Line): This is better than Pure ALOHA. It reaches a higher peak early on because users are forced to send data only at specific times (slots), which reduces the chance of a crash.

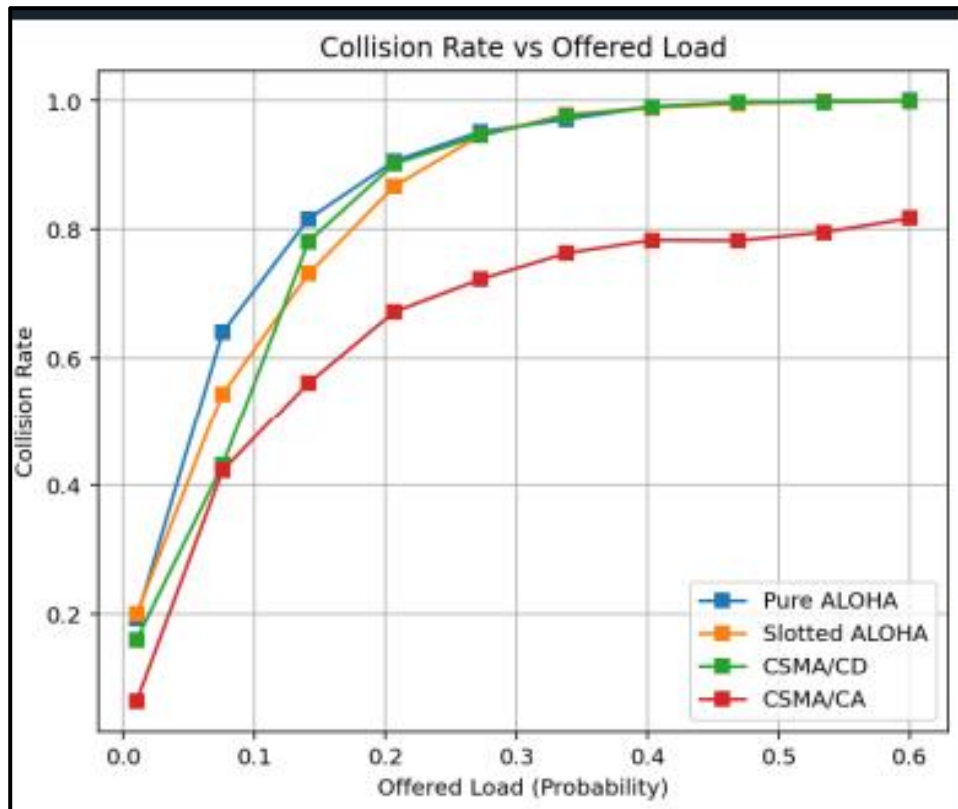
CSMA/CD (Green Line): This protocol "listens" before talking. It performs well at low loads but starts to struggle and drop when the network becomes too crowded.

CSMA/CA (Red Line): This is the **most successful** protocol in the graph. Even when the load is very high (0.6), it maintains the highest throughput. It is the best at avoiding collisions by making users wait for a random time before sending data.

Key Findings:

The results clearly show that **CSMA/CA** is the most efficient protocol for busy networks. **Pure ALOHA** is the least efficient because it cannot handle high traffic without constant collisions.

Graph 2: Collision Rate vs Offered Load



Observations:

General Pattern: As traffic (Load) increases, the number of crashes (collisions) goes up for every protocol. Ideally, we want the collision rate to stay as low as possible for as long as possible.

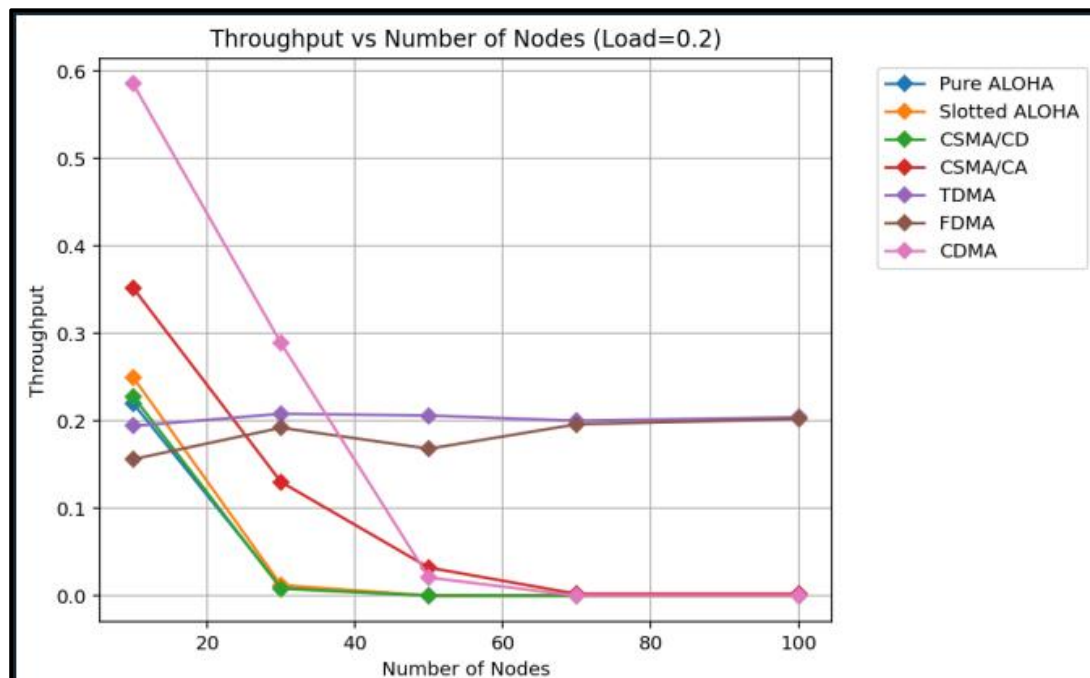
CSMA/CA (Red Line): This is the **most successful** protocol in this graph. Its line stays much lower than the others. Even when the network is very busy (0.6 load), it keeps the collision rate around 0.8 instead of hitting the 1.0 failure mark.

Key Findings:

The results prove that **CSMA/CA** is the best at avoiding crashes. By using a "backoff" rule (making users wait a random time), it prevents the constant "pile-up"

of data that causes the other protocols to fail completely. Pure ALOHA has the highest collision rate. Slotted ALOHA reduces collisions compared to Pure ALOHA.

Graph 3: Throughput vs Number of Nodes



Observations:

General Pattern: As we add more users (Nodes) to the network, the protocols that rely on "randomness" start to fail. However, protocols that are "organized" stay steady and reliable even with 100 users.

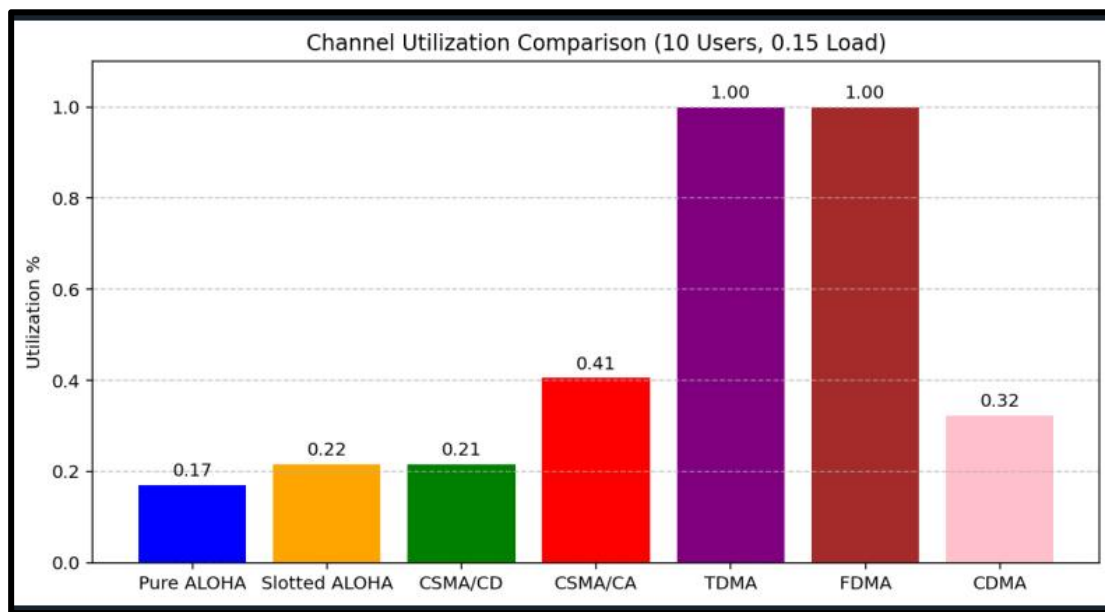
TDMA & FDMA: These are the **most successful** in this graph. Their lines stay flat and steady even when there are 100 nodes. Because they give every user a specific "turn" (TDMA) or "lane" (FDMA), they never have collisions, which makes them very reliable for large groups.

Random Access & CDMA (The falling lines): These protocols cannot handle a large number of users. As the group grows from 10 to 100 nodes, their performance crashes down to zero. This happens because too many people try to talk at the same time, causing constant "accidents" (collisions) that completely block the network.

Key Findings:

The results show that **Scalability** is the main lesson here. **Random Access** protocols (like ALOHA) work fine for a few users but fail in large networks. **Controlled Access** protocols (like TDMA) are the best choice when the number of users is high.

Graph 4: Channel Utilization Comparison



Observations:

General Pattern: This bar chart shows how well each protocol uses the available space on the network. A higher bar means the protocol is very efficient and wastes less time.

Controlled access protocols (TDMA and FDMA) provide the highest channel utilization because they completely avoid collisions by assigning fixed time or frequency resources to users.

CSMA/CA performs significantly better among random access protocols (0.41) as it reduces collisions using a backoff and waiting mechanism before transmission.

CDMA shows moderate utilization (0.32) because multiple users can transmit simultaneously, but performance decreases when network load increases.

Slotted ALOHA (0.22) performs better than **Pure ALOHA** (0.17) since dividing time into slots reduces the probability of collisions.

CSMA/CD (0.21) improves slightly over Pure ALOHA because it detects collisions during transmission and helps reduce wasted channel time.

Key findings:

Overall, random access protocols are less efficient due to collision overhead, while controlled access protocols achieve near-perfect efficiency due to structured channel allocation.

Experimental Scenarios and Analysis

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In [5]: %runfile C:/Users/romai/Desktop/untitled14.py --wdir
```

SECTION 1: STANDARD PROTOCOL PERFORMANCE (Baseline: 10 Users, 0.15 Load)					
Protocol	Thruput	Colis%	Util%	Delay	Fairness
Pure ALOHA	0.012	0.992	0.008	9.993	0.554
Slotted ALOHA	0.352	0.767	0.233	4.085	0.958
CSMA/CD	0.321	0.786	0.214	3.006	0.980
CSMA/CA	0.380	0.606	0.394	4.134	0.977
TDMA	0.150	0.000	0.150	5.000	0.968
FDMA	1.503	0.000	0.150	1.000	0.995
CDMA	1.441	0.018	0.982	1.018	0.994

Base Case Analysis

Pure ALOHA has the worst performance because there are no rules to stop users from sending at the same time, which causes very high collisions.

Slotted ALOHA performs better than Pure ALOHA because it only allows users to transmit at fixed time slots, which reduces collisions.

CSMA/CA is the best random access protocol because it listens to the channel before sending and uses a waiting (backoff) method to avoid collisions.

TDMA and FDMA have zero collisions because each user is given a fixed time slot or frequency, so no overlap occurs.

TDMA removes collisions but causes waiting time since users must wait for their turn even if the channel is free.

FDMA works efficiently but is limited because the number of users depends on how many frequency bands are available.

CDMA allows many users to transmit at the same time using different codes, but performance drops when too many users send data together.

Random access protocols like ALOHA and CSMA are easy to use but become inefficient when the network is heavily loaded.

Controlled access protocols like TDMA and FDMA are more complex but are more stable and reliable for large networks.

CSMA and TDMA show good fairness, meaning all users get a fair chance to transmit data.

When traffic is high, ALOHA suffers from heavy congestion, while CSMA/CA still manages the channel better.

SECTION 2: EXPERIMENTAL SCENARIOS				
Scenario: [Nodes: 10] [Offered Load: 0.1]				
Protocol	Thrput	Colis%	Util%	Delay
Pure ALOHA	0.270	0.743	0.257	7.738
Slotted ALOHA	0.354	0.647	0.353	3.653
CSMA/CD	0.264	0.636	0.364	2.620
CSMA/CA	0.380	0.518	0.482	3.444
TDMA	0.080	0.000	1.000	5.000
FDMA	0.088	0.000	1.000	1.000
CDMA	0.321	0.000	0.333	1.000

Scenario: [Nodes: 10] [Offered Load: 0.5]				
Protocol	Thrput	Colis%	Util%	Delay
Pure ALOHA	0.002	1.000	0.000	9.926
Slotted ALOHA	0.008	0.998	0.002	4.993
CSMA/CD	0.008	0.998	0.002	3.476
CSMA/CA	0.318	0.799	0.201	4.522
TDMA	0.484	0.000	1.000	5.000
FDMA	0.506	0.000	1.000	1.000
CDMA	0.425	0.745	0.085	1.745

Scenario: [Nodes: 50] [Offered Load: 0.1]				
Protocol	Thrput	Colis%	Util%	Delay
Pure ALOHA	0.016	0.997	0.003	9.914
Slotted ALOHA	0.034	0.993	0.007	4.965
CSMA/CD	0.026	0.995	0.005	3.481
CSMA/CA	0.094	0.972	0.028	5.394
TDMA	0.082	0.000	1.000	25.000
FDMA	0.114	0.000	1.000	1.000
CDMA	0.481	0.710	0.097	1.710

Scenario: [Nodes: 50] [Offered Load: 0.5]				
Protocol	Thrput	Colis%	Util%	Delay
Pure ALOHA	0.000	1.000	0.000	9.973
Slotted ALOHA	0.000	1.000	0.000	4.983
CSMA/CD	0.000	1.000	0.000	3.502
CSMA/CA	0.006	0.999	0.001	5.470
TDMA	0.494	0.000	1.000	25.000
FDMA	0.486	0.000	1.000	1.000
CDMA	0.000	1.000	0.000	2.000

Scenario: [Nodes: 100] [Offered Load: 0.1]				
Protocol	Thrput	Colis%	Util%	Delay
Pure ALOHA	0.000	1.000	0.000	9.938
Slotted ALOHA	0.000	1.000	0.000	4.991
CSMA/CD	0.000	1.000	0.000	3.535
CSMA/CA	0.000	1.000	0.000	5.576
TDMA	0.096	0.000	1.000	50.000
FDMA	0.116	0.000	1.000	1.000
CDMA	0.036	0.989	0.004	1.989

Scenario: [Nodes: 100] [Offered Load: 0.5]				
Protocol	Thrput	Colis%	Util%	Delay
Pure ALOHA	0.000	1.000	0.000	10.035
Slotted ALOHA	0.000	1.000	0.000	5.002
CSMA/CD	0.000	1.000	0.000	3.493
CSMA/CA	0.000	1.000	0.000	5.465
TDMA	0.492	0.000	1.000	50.000
FDMA	0.512	0.000	1.000	1.000
CDMA	0.000	1.000	0.000	2.000

Generating Plots... Please check the Plots tab.

1. Effect of Offered Load (Traffic Intensity)

Random Access: At a low load (0.1), these protocols function but still have collisions. However, as the load increases to 0.5, **Pure ALOHA**, **Slotted ALOHA**, and **CSMA/CD** crashes. Only **CSMA/CA** maintains some performance because its "Avoidance" rules prevent total network collapse.

Controlled Access (TDMA & FDMA): These are the most stable. As the load increases, they actually become *more* efficient (Throughput goes up) because they use their assigned slots/lanes more effectively without any risk of collisions.

2. Effect of Number of Nodes (Scalability)

Small Groups (10 Nodes): Most protocols can handle 10 users reasonably well. Even the ALOHA family can send some data successfully.

Large Groups (50–100 Nodes): As the node count grows, random access protocols fails. With 100 nodes, the collision rate hits **1.000 (100%)** for ALOHA and CSMA because too many users are fighting for the same space.

TDMA/FDMA Resilience: These protocols are not affected by node count in terms of collisions. However, in **TDMA**, adding more nodes causes a massive increase in **Delay** (jumping from 5.0 to 50.0) because each user must wait much longer for their specific turn.

3. Comparative Discussion of Protocols

Pure ALOHA vs. Slotted ALOHA: In every scenario, Slotted ALOHA is better. By simply adding a "timing" rule, it cuts collisions and doubles the throughput compared to the "talk anytime" rule of Pure ALOHA.

CSMA/CD vs. CSMA/CA: Listening to the channel (sensing) makes these much better than ALOHA at low loads. However, **CSMA/CA** is superior for wireless simulations; while CD just detects a crash, CA actively waits to avoid it, resulting in higher throughput when the network is busy.

TDMA vs. FDMA: Both provide a "collision-free" environment. **FDMA** is faster (lowest delay), but **TDMA** is more flexible for digital systems. The trade-off is that TDMA becomes very slow (high delay) as more users join the network.

CDMA: It is unique because it allows multiple users to talk at once using codes. It performs very well with 10–50 users at low load but crashes when too many users (100) or high traffic (0.5) cause the signal codes to interfere with each other.

Key Findings:

The simulation proves that **Random Access** is best for simple, quiet networks with few users. For busy or large networks (like 100 nodes), **Controlled Access** or **Smart Sensing (CSMA/CA)** is necessary to prevent the network from crashing.

Conclusion: Key Findings and Insights

The simulation and analysis of different MAC protocols helped us understand how network performance changes with different transmission methods. The results clearly show that the way data is shared in a network has a big impact on efficiency and stability.

Performance Results

Pure ALOHA Weakness:

Pure ALOHA gives the worst performance because there are no rules for coordination. When traffic increases, most packets collide due to a large collision window, so efficiency becomes very low.

Slotted ALOHA Improvement:

Slotted ALOHA performs better than Pure ALOHA because it uses time slots. This reduces the chances of packets overlapping and improves throughput.

CSMA Advantage:

CSMA-based protocols like CSMA/CD and CSMA/CA perform much better because they listen to the channel before sending data. This helps avoid unnecessary collisions and makes them more suitable for busy networks.

TDMA (Time Division):

TDMA avoids collisions completely by giving each user a fixed time slot. The drawback is that users still have to wait for their turn even if the channel is free.

FDMA (Frequency Division):

FDMA also avoids collisions by assigning separate frequency channels to users. However, the number of users is limited because each user needs a separate frequency band.

CDMA (Code Division):

CDMA allows multiple users to transmit at the same time using different codes. But when too many users are active, interference increases and performance starts to drop.

Main Trade-offs

Efficiency vs Complexity:

Simple protocols like ALOHA are easy to use but waste more bandwidth. More advanced protocols like CSMA/CA and CDMA use the network more efficiently but are more complex to implement.

Delay vs Scalability:

Random access protocols work well when traffic is low, but they fail when the network becomes busy. Controlled access protocols like TDMA handle large numbers of users better, but they introduce fixed waiting time.

Final Conclusion

This simulation shows that no single MAC protocol is perfect for all situations. Random access methods are useful for small or low-traffic networks, while controlled access and CSMA-based methods are better for large and busy networks where reliability and fewer collisions are important.

References

Lecture notes and slides provided by course instructor

IEEE 802.11 Standard documentation (for CSMA/CA understanding)

Online study resources for MAC protocols (GeeksforGeeks, TutorialsPoint)

Wireless Communications and Networking – William Stallings

Results generated via the custom Python environment using parameters for 10, 50, and 100 nodes at varying offered loads (0.1 to 0.5).